

FagusED · A-Level 9231
1.1 Roots of polynomial equations

What is the central idea in Roots of polynomial equations?

For polynomial equations, Vieta's formulas link symmetric sums/products of roots to coefficients. If coefficients are real, non-real complex roots occur in conjugate pairs.

9231-01-C1

QUESTION + ANSWER

FagusED · A-Level 9231
1.1 Roots of polynomial equations

Which mechanism, condition or distinction matters in Roots of polynomial equations?

Transformed roots can be found without solving the original polynomial by substituting a new variable and clearing denominators carefully.

9231-01-C2

QUESTION + ANSWER

FagusED · A-Level 9231
1.1 Roots of polynomial equations

A monic cubic has roots 1, 2 and 4. Write equation whose roots are their reciprocals.

Reciprocal roots 1, $1/2$, $1/4$; multiply $(x-1)(2x-1)(4x-1)$; $8x^3 - 14x^2 + 7x - 1 = 0$.

9231-01-P

QUESTION + ANSWER

FagusED · A-Level 9231
1.2 Rational functions and graphs

Which mechanism, condition or distinction matters in Rational functions and graphs?

Differentiation locates stationary points; sign charts show intervals where the curve rises and falls. A cancelled factor may produce a hole rather than a vertical asymptote.

9231-02-C2

QUESTION + ANSWER

FagusED · A-Level 9231
1.2 Rational functions and graphs

Find vertical and horizontal asymptotes of $y=(2x+1)/(x-3)$ and identify the x-intercept.

Denominator zero gives $x=3$; ratio leading coefficients $y=2$; numerator zero gives $x=-1/2$.

9231-02-P

QUESTION + ANSWER

FagusED · A-Level 9231
1.2 Rational functions and graphs

What is the central idea in Rational functions and graphs?

A rational graph is controlled by zeros of numerator, excluded denominator values, asymptotes and possible cancellations. Divide polynomials for an oblique asymptote.

9231-02-C1

QUESTION + ANSWER

FagusED · A-Level 9231
1.3 Summation of series

Find Σ from $r=1$ to n of $1/[r(r+1)]$.

Partial fractions $1/[r(r+1)] = 1/r - 1/(r+1)$; cancellation; sum $= 1 - 1/(n+1) = n/(n+1)$.

9231-03-P

QUESTION + ANSWER

FagusED · A-Level 9231
1.3 Summation of series

Which mechanism, condition or distinction matters in Summation of series?

Induction is useful to prove a proposed result: verify base case, assume $n=k$, then derive $n=k+1$ explicitly. An infinite geometric sum requires $\text{abs}(r) < 1$; never apply it to divergent series.

9231-03-C2

QUESTION + ANSWER

FagusED · A-Level 9231
1.3 Summation of series

What is the central idea in Summation of series?

Use finite-sum identities for r , r^2 , r^3 and partial fractions for telescoping sums.

9231-03-C1

QUESTION + ANSWER

FagusED · A-Level 9231
1.4 Matrices

Which mechanism, condition or distinction matters in Matrices?

Eigenvectors retain direction under transformation, with eigenvalue scale factor. An inverse exists only when determinant is nonzero.

9231-04-C2

QUESTION + ANSWER

FagusED · A-Level 9231
1.4 Matrices

For $A=[[2,0],[0,-1]]$, find eigenvalues and determinant, and describe transformation of x and y coordinates.

Eigenvalues 2 and -1 ; $\det A = -2$; x doubles while y is reflected across x -axis.

9231-04-P

QUESTION + ANSWER

FagusED · A-Level 9231
1.4 Matrices

What is the central idea in Matrices?

A matrix represents a linear transformation. Matrix multiplication composes transformations in reverse order of application; determinants reveal area scale and invertibility.

9231-04-C1

QUESTION + ANSWER

FagusED · A-Level 9231
1.5 Polar coordinates

Which mechanism, condition or distinction matters in Polar coordinates?

For a polar curve, area between $\theta=a$ and b is $\frac{1}{2}\int_a^b r^2 d\theta$; intersections may require considering repeated representations of the same point.

9231-05-C2

QUESTION + ANSWER

FagusED · A-Level 9231
1.5 Polar coordinates

Find area swept by polar curve $r=2$ from $\theta=0$ to $\theta=\pi/2$.

$A=\frac{1}{2}\int_0^{\pi/2} r^2 d\theta = \frac{1}{2} \times 4 \times \pi/2 = \pi$ square units.

9231-05-P

QUESTION + ANSWER

FagusED · A-Level 9231
1.5 Polar coordinates

What is the central idea in Polar coordinates?

Polar coordinates use $x=r \cos \theta$, $y=r \sin \theta$. Negative r represents a point in the opposite direction to θ , so specify conventions carefully.

9231-05-C1

QUESTION + ANSWER

What is the central idea in Vectors?

A vector line can be $r=a+\lambda b$. To find intersections, equate components and solve for distinct parameters. A plane $r \cdot n=d$ has normal vector n .

9231-06-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Proof by induction?

The inductive hypothesis is used at a visible algebraic step, not merely asserted. For divisibility, rewrite the $n=k+1$ expression as a known divisible term plus a clearly divisible remainder.

9231-07-C2

QUESTION + ANSWER

Show $\cosh^2 x - \sinh^2 x = 1$ directly from exponential definitions.

Difference of squares:
 $((e^x + e^{-x})^2 - (e^x - e^{-x})^2)/4 = 4e^x e^{-x}/4 = 1$.

9231-08-P

QUESTION + ANSWER

For $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$, give eigenvalues and one eigenvector for each.

Eigenvalues 3 and 5; $v_3 = (1, 0)$, $v_5 = (0, 1)$; nonzero scalar multiples accepted.

9231-09-P

QUESTION + ANSWER

Find dy/dx on $x^2 + y^2 = 25$ at $(3, 4)$ and state tangent gradient.

Differentiate: $2x + 2y y' = 0$; $y' = -x/y$; at $(3, 4)$, $-3/4$.

9231-10-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Vectors?

Dot product tests perpendicularity and angles; cross product gives a perpendicular vector and parallelogram area.

9231-06-C2

QUESTION + ANSWER

Prove $1+3+\dots+(2n-1)=n^2$ by induction.

$n=1$: $1=1^2$. Assume sum through k is k^2 ; add $2(k+1)-1=2k+1$ to get $k^2+2k+1=(k+1)^2$.

9231-07-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Hyperbolic functions?

Inverse hyperbolic functions can be written using logarithms by solving a quadratic in e^x .

9231-08-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Matrices?

Solve $\det(A - \lambda I) = 0$ for eigenvalues; an eigenvector satisfies $(A - \lambda I)v = 0$. Repeated eigenvalues need enough independent eigenvectors for diagonalisation.

9231-09-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Differentiation?

For polar curves, compute derivatives through $x=r \cos \theta$ and $y=r \sin \theta$. Higher derivatives locate turning-point type and curvature.

9231-10-C2

QUESTION + ANSWER

Find angle between $(1, 0, 0)$ and $(1, 1, 0)$, giving exact value.

Dot product = 1; lengths 1 and $\sqrt{2}$; $\cos \theta = 1/\sqrt{2}$, $\theta = \pi/4$.

9231-06-P

QUESTION + ANSWER

What is the central idea in Proof by induction?

Induction proves a proposition for all positive integers by establishing a base case and showing $P(k)$ implies $P(k+1)$.

9231-07-C1

QUESTION + ANSWER

What is the central idea in Hyperbolic functions?

$\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$. Their identity $\cosh^2 x - \sinh^2 x = 1$ contrasts with trigonometric identities; derivatives are $\cosh$ and $\sinh$ respectively.

9231-08-C1

QUESTION + ANSWER

What is the central idea in Matrices?

A Level matrix work extends to three-dimensional transformations, eigenvalues and diagonalisation where applicable.

9231-09-C1

QUESTION + ANSWER

What is the central idea in Differentiation?

Implicit differentiation treats y as a function of x ; derivatives of y terms acquire dy/dx . Parametric differentiation uses $dy/dx = (dy/dt)/(dx/dt)$, provided $dx/dt \neq 0$.

9231-10-C1

QUESTION + ANSWER

2.4 Integration

What is the central idea in Integration?

Integration may require substitution, partial fractions, integration by parts or reduction formulae. Check domains for logarithmic antiderivatives; $\int 1/x \, dx = \ln|x| + C$.

9231-11-C1

[QUESTION + ANSWER](#)

2.5 Complex numbers

Find the two square roots of -4 in complex numbers and give their moduli.

$z^2 = -4$ gives $z = \pm 2i$; each modulus is 2.

9231-12-P

[QUESTION + ANSWER](#)

2.6 Differential equations

Which mechanism, condition or distinction matters in Differential equations?

Particular solutions require applying the initial condition after finding the general solution; an omitted constant loses a family of solutions.

9231-13-C2

[QUESTION + ANSWER](#)

3.1 Motion of a projectile

A particle is projected at $25 \, \text{m s}^{-1}$ at 30° above horizontal. Find maximum height, $g = 10 \, \text{m s}^{-2}$.

$u_y = 25 \sin 30^\circ = 12.5$; at top $v_y = 0$, $h = u_y^2 / (2g) = 7.8125 \, \text{m}$.

9231-14-P

[QUESTION + ANSWER](#)

3.2 Equilibrium of a rigid body

A horizontal uniform beam 4 m long has weight 80 N at centre; a support at one end and upward force at other hold it. Find upward force at far end.

Take moments about support: $F \times 4 = 80 \times 2$; $F = 40 \, \text{N}$.

9231-15-P

[QUESTION + ANSWER](#)

2.4 Integration

Which mechanism, condition or distinction matters in Integration?

Definite integrals can represent signed area, so use absolute value or split where actual geometric area is asked.

9231-11-C2

[QUESTION + ANSWER](#)

2.5 Complex numbers

Which mechanism, condition or distinction matters in Complex numbers?

Conjugation reflects across real axis. Roots of a complex number lie evenly spaced around a circle; De Moivre's theorem gives powers.

9231-12-C2

[QUESTION + ANSWER](#)

2.6 Differential equations

Solve $dy/dx = 3y$ given $y(0) = 2$.

Separate $dy/y = 3dx$; $\ln|y| = 3x + C$; $y = 2e^{3x}$.

9231-13-P

[QUESTION + ANSWER](#)

3.1 Motion of a projectile

Which mechanism, condition or distinction matters in Motion of a projectile?

Use common time to combine components; at maximum height vertical velocity is zero. Range at equal launch/landing height is $u^2 \sin(2\theta)/g$.

9231-14-C2

[QUESTION + ANSWER](#)

3.2 Equilibrium of a rigid body

Which mechanism, condition or distinction matters in Equilibrium of a rigid body?

Take moments about a point that removes unknown forces to solve efficiently, then check force components and physical sign.

9231-15-C2

[QUESTION + ANSWER](#)

2.4 Integration

Evaluate $\int$ from 0 to 1 of $x e^x \, dx$.

Parts gives antiderivative $e^x(x-1)$; evaluate at 1 minus at 0: $0 - (-1) = 1$.

9231-11-P

[QUESTION + ANSWER](#)

2.5 Complex numbers

What is the central idea in Complex numbers?

Write $z = x + iy$, modulus $\sqrt{x^2 + y^2}$, argument with quadrant check; polar form $r(\cos \theta + i \sin \theta)$ simplifies multiplication and powers.

9231-12-C1

[QUESTION + ANSWER](#)

2.6 Differential equations

What is the central idea in Differential equations?

Separate variables only when dy/dx can be arranged as $f(y)g(x)$. For linear first-order equations use an integrating factor when required.

9231-13-C1

[QUESTION + ANSWER](#)

3.1 Motion of a projectile

What is the central idea in Motion of a projectile?

Resolve launch velocity into perpendicular components. Horizontal acceleration is zero and vertical acceleration is $-g$ when air resistance is ignored.

9231-14-C1

[QUESTION + ANSWER](#)

3.2 Equilibrium of a rigid body

What is the central idea in Equilibrium of a rigid body?

Rigid-body equilibrium requires both resultant force zero and resultant moment zero. Weight acts through centre of mass; contact reactions and friction depend on geometry.

9231-15-C1

[QUESTION + ANSWER](#)

FagusED · A-Level 9231
3.3 Circular motion

What is the central idea in Circular motion?

For circular movement, acceleration points inward with magnitude $v^2/r = \omega^2 r$. In vertical circles, radial resultant changes with position because the radial component of weight changes.

9231-16-C1

QUESTION + ANSWER

FagusED · A-Level 9231
3.3 Circular motion

Which mechanism, condition or distinction matters in Circular motion?

State the chosen positive radial direction before writing Newton's second law.

9231-16-C2

QUESTION + ANSWER

FagusED · A-Level 9231
3.3 Circular motion

A 0.5 kg mass travels at 4 m s^{-1} around radius 2 m on a horizontal frictionless table. Determine string tension.

Radial acceleration $v^2/r = 16/2 = 8 \text{ m s}^{-2}$; tension supplies resultant; $T = ma = 4 \text{ N}$.

9231-16-P

QUESTION + ANSWER

FagusED · A-Level 9231
3.4 Hooke's law

Which mechanism, condition or distinction matters in Hooke's law?

For systems with two springs, identify whether common force (series) or common extension (parallel) applies before replacing them by an effective stiffness.

9231-17-C2

QUESTION + ANSWER

FagusED · A-Level 9231
3.4 Hooke's law

What is the central idea in Hooke's law?

A Hookean spring obeys $T = kx$ for extension x within elastic limit. Elastic energy stored is $\frac{1}{2}kx^2$, the area under its force-extension graph.

9231-17-C1

QUESTION + ANSWER

FagusED · A-Level 9231
3.4 Hooke's law

A spring of stiffness 200 N m^{-1} stretches 0.10 m. Calculate tension and energy stored.

$T = kx = 20 \text{ N}$; energy $= \frac{1}{2}kx^2 = 1.0 \text{ J}$.

9231-17-P

QUESTION + ANSWER

FagusED · A-Level 9231
3.5 Linear motion under a variable force

A 2 kg particle at rest is pushed by $F = 6x \text{ N}$ from $x = 0$ to $x = 2 \text{ m}$. Find speed at $x = 2 \text{ m}$.

Work $\int_0^2 6x \, dx = 12 \text{ J}$; $\frac{1}{2}(2)v^2 = 12$, $v = \sqrt{12} \approx 3.46 \text{ m s}^{-1}$.

9231-18-P

QUESTION + ANSWER

FagusED · A-Level 9231
3.5 Linear motion under a variable force

Which mechanism, condition or distinction matters in Linear motion under a variable force?

The work-energy method $\int F \, dx = \Delta(\frac{1}{2}mv^2)$ can simplify finding speed without time.

9231-18-C2

QUESTION + ANSWER

FagusED · A-Level 9231
3.5 Linear motion under a variable force

What is the central idea in Linear motion under a variable force?

For variable force, Newton's law is $m \, dv/dt = F(t, x, v)$. If acceleration is a function of position, use $a = v \, dv/dx$, then integrate.

9231-18-C1

QUESTION + ANSWER

FagusED · A-Level 9231
3.6 Momentum

A 1 kg trolley at 4 m s^{-1} hits a stationary 1 kg trolley; they stick. Find common velocity and kinetic-energy change.

Momentum gives $(1)(4) = (2)v$, so $v = 2 \text{ m s}^{-1}$; initial KE 8 J, final 4 J, change -4 J .

9231-19-P

QUESTION + ANSWER

FagusED · A-Level 9231
3.6 Momentum

Which mechanism, condition or distinction matters in Momentum?

The coefficient of restitution e compares relative separation speed to approach speed along the line of impact, ranging from 0 to 1 in usual collision models.

9231-19-C2

QUESTION + ANSWER

FagusED · A-Level 9231
3.6 Momentum

What is the central idea in Momentum?

Impulse is $\int F \, dt = \text{change in momentum}$. Momentum is conserved for an isolated system; kinetic energy is conserved only in elastic collisions.

9231-19-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.1 Continuous random variables

For $f(x) = kx$ on $0 \leq x \leq 2$, zero elsewhere, find k and $P(X > 1)$.

$1 = \int_0^2 kx \, dx = 2k$, so $k = 1/2$;
 $P(X > 1) = \int_1^2 x/2 \, dx = 3/4$.

9231-20-P

QUESTION + ANSWER

FagusED · A-Level 9231
4.1 Continuous random variables

Which mechanism, condition or distinction matters in Continuous random variables?

Find $E(X) = \int x f(x) \, dx$ and $\text{Var}(X) = E(X^2) - E(X)^2$; transform a distribution by checking domains and Jacobian when needed.

9231-20-C2

QUESTION + ANSWER

FagusED · A-Level 9231
4.1 Continuous random variables

What is the central idea in Continuous random variables?

A probability density integrates to one and probabilities are areas, not heights. For continuous X , $P(X = a) = 0$.

9231-20-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.2 Inference using normal and t-distributi...

What is the central idea in Inference using normal and t-distributions?

A confidence interval estimates a population parameter with stated coverage procedure.

9231-21-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.3 Chi-square tests

Which mechanism, condition or distinction matters in Chi-square tests?

Association tests use expected count row total $\times$ column total/grand total.

9231-22-C2

QUESTION + ANSWER

FagusED · A-Level 9231
4.4 Non-parametric tests

What is the central idea in Non-parametric tests?

Rank-based tests use order rather than a specified normal distribution. A sign test checks direction of paired changes after discarding zero differences; under null, plus signs follow Binomial(n , 0.5).

9231-23-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.5 Probability generating functions

What is the central idea in Probability generating functions?

A probability generating function $G_X(t) = E(t^X) = \sum p_r t^r$ for non-negative integer X .

9231-24-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.2 Inference using normal and t-distributi...

Which mechanism, condition or distinction matters in Inference using normal and t-distributions?

For known σ and appropriate normal conditions use z ; with unknown σ and small normal sample use Student's t with $n-1$ degrees of freedom. State null/alternative, assumptions, test statistic and conclusion in context.

9231-21-C2

QUESTION + ANSWER

FagusED · A-Level 9231
4.3 Chi-square tests

Observed counts 30 and 20; expected 25 and 25. Calculate χ^2 and degrees of freedom for two categories with no fitted parameters.

$$\chi^2 = (30-25)^2/25 + (20-25)^2/25 = 2.0; \\ df = 2-1 = 1.$$

9231-22-P

QUESTION + ANSWER

FagusED · A-Level 9231
4.4 Non-parametric tests

In six paired observations, five differences are positive and one negative. Give two-sided sign-test p -value under H_0 .

$$P(X \geq 5) = C(6,5)/64 + C(6,6)/64 = 7/64; \\ \text{two-sided } p = 14/64 = 0.21875.$$

9231-23-P

QUESTION + ANSWER

FagusED · A-Level 9231
4.5 Probability generating functions

A variable has $P(X=0)=0.3$ and $P(X=1)=0.7$. Find PGF and $E(X)$ using derivative.

$$G(t) = 0.3 + 0.7t; \quad G'(t) = 0.7; \\ E(X) = G'(1) = 0.7.$$

9231-24-P

QUESTION + ANSWER

FagusED · A-Level 9231
4.2 Inference using normal and t-distributi...

For $n=16$, sample $s=4$ and mean=50, calculate standard error and degrees of freedom for a t interval.

$$s/\sqrt{n} = 4/4 = 1; \text{ degrees of freedom } 15.$$

9231-21-P

QUESTION + ANSWER

FagusED · A-Level 9231
4.3 Chi-square tests

What is the central idea in Chi-square tests?

A chi-square goodness-of-fit statistic sums $(O-E)^2/E$ across categories. Expected counts must be large enough for approximation; pool categories where needed and adjust degrees of freedom for constraints/estimated parameters.

9231-22-C1

QUESTION + ANSWER

FagusED · A-Level 9231
4.4 Non-parametric tests

Which mechanism, condition or distinction matters in Non-parametric tests?

Wilcoxon signed-rank additionally uses magnitudes via ranks, subject to its assumptions.

9231-23-C2

QUESTION + ANSWER

FagusED · A-Level 9231
4.5 Probability generating functions

Which mechanism, condition or distinction matters in Probability generating functions?

$G(1)=1$, $G'(1)=E(X)$, and $G''(1)+G'(1)-G'(1)^2=\text{Var}(X)$. Independent sums multiply PGFs, allowing convolution without listing every outcome.

9231-24-C2

QUESTION + ANSWER